

Supplementary Information:
Message-capacity limits of individual agents set the transition
point of collective truth-finding in language-model networks

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1 Reanalysis of the public dataset

1.1 Endogeneity of the initial fraction

Variance decomposition of x_0 and within/between-question outcome correlations (random-graph family, $\delta = 0.1$), computed over the 6,400 objective communities of El et al.:

model	between-question share of $\text{var}(x_0)$	$\text{corr}(x_0, \text{outcome})$	between	within
Gemma-3n	0.995	0.802	0.852	0.041
Llama-3-8B	0.936	0.659	0.721	0.162
Qwen3.5-9B	0.861	0.277	0.351	0.132
GPT-4o-mini	0.503	0.298	0.321	0.273

Question-clustered versus i.i.d. bootstrap intervals for the fitted $q(x_0)$ crossing (random family):

model	crossing	i.i.d. CI	width	question-clustered CI	ratio
GPT-4o-mini	0.494	[0.477, 0.513]	0.036	[0.404, 0.583]	$4.9\times$
Gemma-3n	0.285	[0.251, 0.315]	0.065	[0.163, 0.419]	$3.9\times$
Qwen3.5-9B	0.142	[0.059, 0.206]	0.147	[-1.049, 0.367]	$9.6\times$
Llama-3-8B	0.459	[0.441, 0.480]	0.039	[0.370, 0.564]	$5.0\times$

1.2 Message-level coefficients and the truth asymmetry

Question-clustered 95% intervals for the per-message truth asymmetry w_T (43,500 observations per model):

model	w_T	cluster 95% CI	Ising-coupling estimate	contains 1
GPT-4o-mini	1.055	[0.996, 1.114]	1.16	yes (boundary)
Gemma-3n	1.179	[0.731, 1.628]	1.71	yes
Qwen3.5-9B	0.900	[0.759, 1.041]	1.33	yes
Llama-3-8B	1.144	[0.972, 1.316]	1.38	yes

The label attenuation λ_{lab} (per-message weight of “disagree”-labeled edges relative to “agree”-labeled ones) is 0.080 ± 0.044 (GPT-4o-mini), 0.555 ± 0.078 (Gemma), 0.294 ± 0.051 (Qwen), 0.288 ± 0.041 (Llama); θ is nonzero in all four models ($t \geq 3.4$, question-clustered).

2 Single-agent measurement (llama3.1:8b)

2.1 Response coefficients

variant	coefficient	estimate	95% CI (claim cluster)	n_{obs}
A	c_0	0.725	[0.516, 0.915]	10608
A	h	0.725	[0.516, 0.915]	10608
A	β_T	0.419	[0.353, 0.486]	10608
A	β_F	-0.568	[-0.701, -0.446]	10608
A	w_T	0.738	[0.603, 0.899]	10608
B	c_0	-0.758	[-0.895, -0.627]	21216
B	h	0.196	[0.062, 0.309]	21216
B	θ	1.909	[1.723, 2.106]	21216
B	β_T	0.291	[0.267, 0.319]	21216
B	β_F	-0.326	[-0.374, -0.287]	21216
B	w_T	0.894	[0.801, 0.976]	21216

2.2 Attenuation-form comparison

scheme	parameter	log-lik	AIC	CV log-lik/obs
load_decay	0.14	-14901.6	29845.3	-0.48256
divisive	0.9	-14781.8	29605.6	-0.48276
pos_decay	0.25	-15164.0	30370.1	-0.48871
topk	5.0	-15470.0	30982.0	-0.49778
null (additive)	0.0	-15807.9	31655.8	-0.50911

Per- k coefficients (variant A): $\beta_T(k) = 3.074, 1.706, 1.126, 0.640, 0.450, 0.327$ and $\beta_F(k) = -0.779, -0.763, -0.743, -0.537, -0.416, -0.331$ at $k = 1, 2, 3, 5, 8, 12$; $k \beta_T(k) \approx 3.1$ – 3.9 (saturation).

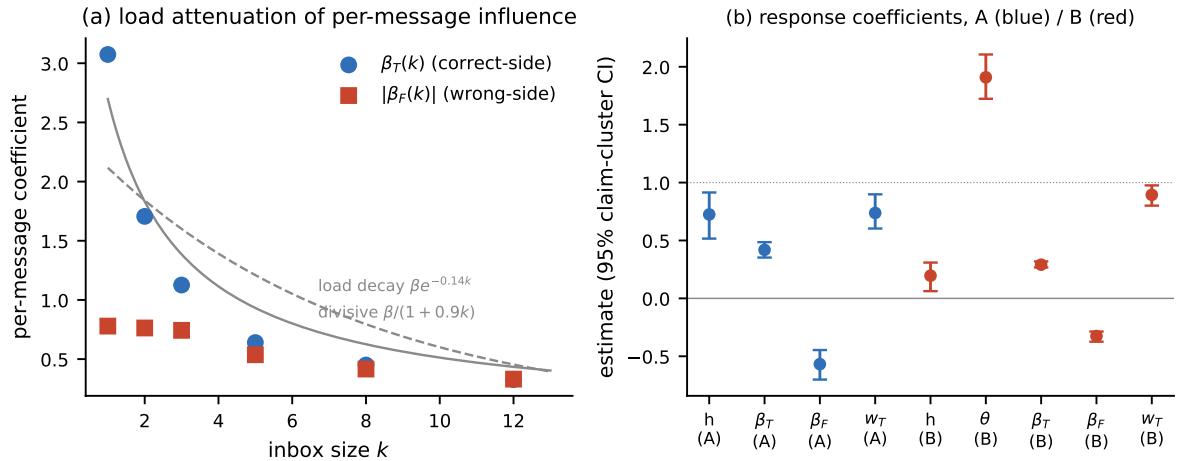


Figure S1: **Per-message coefficients versus inbox size, and coefficient estimates for prompt variants A and B (31,824 randomized queries).** (a) Per-message coefficients attenuate with inbox size k ; the two inbox-size-dependent forms (gray) beat position-dependent decay in claim-clustered cross-validation. (b) Coefficient estimates with 95% claim-clustered bootstrap intervals for prompt variants A (no own-state line) and B (explicit own-state line).

3 Closed-form α^* : validation table

Exact α^* (full $P(k)$ sum) versus the two closure levels, for seven synthetic coefficient sets and the four measured ones (notes/08):

parameter set	exact α^*	G-fold (exact $E[K]$)	fully closed
$h=.10, \theta=.6, \bar{\beta}=.5$	0.62	0.640	0.593
$h=.10, \theta=.6, \bar{\beta}=1$	1.40	1.415	1.248
$h=.30, \theta=.6, \bar{\beta}=1$	1.12	1.172	1.048
$h=.70, \theta=.6, \bar{\beta}=1$	0.84	0.882	0.804
$h=.10, \theta=0, \bar{\beta}=1$	1.18	1.199	1.070
$h=.10, \theta=2, \bar{\beta}=.5$	0.90	0.899	0.818
$h=.30, \theta=2, \bar{\beta}=1$	1.54	1.568	1.372
variant A, measured (constant)	0.42	0.435 ($K^*=6.4$)	0.409
variant B, measured (constant)	0.46	0.470 ($K^*=6.1$)	0.440
variant A, measured (divisive)	0.46	0.502 ($K^*=5.9$)	0.469
variant B, measured (divisive)	1.26	1.284 ($K^*=3.2$)	1.141

Median relative error: 2.1% (exact- $E[K]$ inversion), 6.4% (fully closed); worst case 9% (variant-A divisive).

4 Main collective experiment

4.1 Committor family (arm A, generated graphs)

α	$32x_0$	n	q	Wilson 95%	undec.	$q_{\delta=.05}$	$q_{\delta=.2}$	q_{excl}
0.2	4	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.2	8	40	0.025	[0.004, 0.129]	0.000	0.025	0.000	0.025
0.2	12	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.2	16	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.2	20	40	0.100	[0.040, 0.231]	0.000	0.100	0.100	0.100
0.2	24	40	0.275	[0.161, 0.428]	0.000	0.275	0.275	0.275
0.3	4	40	0.000	[-0.000, 0.088]	0.025	0.000	0.000	0.000
0.3	8	16	0.000	[0.000, 0.194]	0.062	0.062	0.000	0.000
0.3	12	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.3	16	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.3	20	40	0.075	[0.026, 0.199]	0.200	0.100	0.050	0.094
0.3	24	40	0.400	[0.263, 0.554]	0.075	0.425	0.400	0.432
0.38	4	16	0.000	[0.000, 0.194]	0.000	0.000	0.000	0.000
0.38	8	16	0.000	[0.000, 0.194]	0.062	0.000	0.000	0.000
0.38	12	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.38	16	40	0.025	[0.004, 0.129]	0.075	0.025	0.000	0.027
0.38	20	40	0.150	[0.071, 0.291]	0.050	0.200	0.150	0.158
0.38	24	40	0.350	[0.221, 0.505]	0.125	0.350	0.325	0.400
0.45	4	40	0.000	[-0.000, 0.088]	0.025	0.000	0.000	0.000
0.45	8	40	0.000	[-0.000, 0.088]	0.150	0.000	0.000	0.000
0.45	12	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.45	16	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.45	20	16	0.000	[0.000, 0.194]	0.312	0.062	0.000	0.000
0.45	24	40	0.450	[0.307, 0.602]	0.075	0.475	0.450	0.486
0.55	4	16	0.000	[0.000, 0.194]	0.062	0.000	0.000	0.000
0.55	8	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.55	12	16	0.000	[0.000, 0.194]	0.125	0.000	0.000	0.000
0.55	16	40	0.100	[0.040, 0.231]	0.175	0.125	0.050	0.121
0.55	20	40	0.225	[0.123, 0.375]	0.175	0.225	0.150	0.273
0.55	24	40	0.375	[0.242, 0.530]	0.075	0.400	0.300	0.405
1	4	16	0.125	[0.035, 0.360]	0.188	0.188	0.125	0.154
1	8	16	0.062	[0.011, 0.283]	0.125	0.125	0.000	0.071
1	12	16	0.062	[0.011, 0.283]	0.312	0.125	0.000	0.091
1	16	40	0.225	[0.123, 0.375]	0.275	0.300	0.150	0.310
1	20	40	0.425	[0.285, 0.578]	0.125	0.450	0.300	0.486
1	24	40	0.400	[0.263, 0.554]	0.275	0.525	0.325	0.552

4.2 Empirical critical crowding level

α	n (episodes, $x_0 \leq 8/32$)	wrong-consensus rate	claim-cluster 95% CI
0.2	56	0.982	[0.946, 1.000]
0.3	56	0.964	[0.893, 1.000]
0.38	32	0.969	[0.906, 1.000]
0.45	80	0.912	[0.775, 1.000]
0.55	32	0.906	[0.719, 1.000]
1	32	0.750	[0.438, 1.000]

No crossing of the pre-registered 20% threshold occurs in the window (α_{emp}^* undefined; 0 of 2,000 claim-clustered bootstrap resamples cross).

4.3 Discrimination arm (variant B)

α	n	wrong rate ($x_0 \leq 10/32$)	95% CI	wrong basin retained
0.6	40	0.975	[0.925, 1.000]	True
0.9	40	1.000	[1.000, 1.000]	True
1.3	40	0.975	[0.925, 1.000]	True

With claim-resolved coefficients, predicted versus observed wrong-basin attractor positions:

α	constant rule x^*	normalized rule x^*	observed low- x_0 $\bar{x}_8$
0.60	0.157	0.051	0.028–0.095
0.90	0.338	0.067	0.061–0.072
1.30	0.438	0.093	0.092–0.097

4.4 Rewiring control

$32x_0$	$n_{\rho=0}$	$q_{\rho=0}$	$n_{\rho=1}$	$q_{\rho=1}$	Δq
4	40	0.000	16	0.000	+0.0000
8	16	0.000	16	0.000	+0.0000
12	16	0.000	16	0.000	+0.0000
16	16	0.000	16	0.000	+0.0000
20	40	0.075	16	0.062	-0.0125
24	40	0.400	16	0.312	-0.0875

4.5 Yoked one-step calibration

α	transitions	Brier	Brier (climatology)	ECE
0.2	43008	0.1028		0.1735 0.0519
0.3	67584	0.1149		0.1931 0.0591
0.38	43008	0.1298		0.2123 0.0757
0.45	43008	0.1193		0.1984 0.0629
0.55	43008	0.1384		0.2247 0.0851
1	43008	0.1581		0.2465 0.1014

The systematic error concentrates at contested compositions: at $l/k \in [0.4, 0.6]$ the pooled rule predicts $P(\text{correct}) = 0.574$ against an observed 0.390 ($n = 17,047$), a -0.20 wrong-ward bias present identically at measured and interpolated k and at all α .

5 Polarity discrimination experiment

5.1 SUPPORTS-claim collectives (A-1)

claim	α	$32x_0 = 4$	$32x_0 = 12$	$32x_0 = 20$	$32x_0 = 28$
993	0.3	0.820	1.000	1.000	1.000
993	0.55	0.974	1.000	1.000	1.000
1103	0.3	0.833	0.875	0.898	0.919
1103	0.55	0.667	0.766	0.784	0.792
1239	0.3	1.000	1.000	1.000	1.000
1239	0.55	1.000	1.000	1.000	1.000

5.2 Negated-rewrite collectives (A-2) versus originals

rewrite	original	$32x_0 = 4$	$32x_0 = 12$	$32x_0 = 20$	$32x_0 = 28$	orig. mean $\bar{x}_8$ ($x_0 \leq 20/32$)
90006	6	0.984	1.000	1.000	1.000	0.282
90382	382	1.000	1.000	1.000	1.000	0.001
91569	1569	1.000	1.000	1.000	1.000	0.223
92070	2070	1.000	1.000	1.000	1.000	0.029

5.3 Claim texts, rewrites, and checks

Selected SUPPORTS claims (bare-prompt p_q in parentheses): 993 (0.708) “While there has been a mean rise of a little more than 3mm per year worldwide since the 1990s, in the last decade, the NOAA Virginia Key tide gauge just south of Miami Beach has measured a 9mm rise annually.”; 1103 (0.792) “[ocean acidification was] First referenced in a peer-reviewed study in Nature in 2003”; 1239 (0.792) “Through decades of research, it has become clear that human civilization, roughly 6,000 years old, developed during an unusually stable period for global sea levels.”

Negated rewrites (model opposition-judgment rate; bare-prompt p_q of the rewrite, TRUE = correct side):

original	rewrite	opp. rate	p_q
The polar bear population has been growing.	The polar bear population has <i>not</i> been growing.	0.70	0.344
Sea level rise is decelerating.	Sea level rise is <i>not</i> decelerating.	0.85	0.938
The amount of energy used to construct solar and wind facilities is greater than they produce in their working lives.	...is <i>not</i> greater than they produce in their working lives.	0.70	0.938
When life is considered, ocean acidification is often found to be a non-problem, or even a benefit.	...is often found to be a <i>problem</i> , and <i>not a benefit</i> (De Morgan transformation).	0.85	1.000

6 Cross-model campaigns

6.1 qwen3:8b: frozen predictions versus observed collectives

claim	α	c_0	frozen fixed points	observed $\bar{x}_8$ ($32x_0 = 4/12/20/28$)
6	0.3	0.418	0.0(stable); 0.149(unstable)	0.200 / 1.000 / 1.000 / 1.000
6	0.55	0.418	0.001(stable); 0.121(unstable); 1.0(stable)	0.325 / 1.000 / 1.000 / 1.000
6	1	0.418	0.003(stable); 0.043(unstable)	0.884 / 1.000 / 1.000 / 1.000
1239	0.3	0.401	0.0(stable); 0.661(unstable); 1.0(stable)	0.000 / 0.000 / 0.000 / 1.000
1239	0.55	0.401	0.0(stable); 0.653(unstable); 1.0(stable)	0.000 / 0.000 / 0.166 / 1.000
1239	1	0.401	0.0(stable); 0.621(unstable); 1.0(stable)	0.000 / 0.000 / 0.422 / 0.944
1569	0.3	-0.279	0.0(stable); 0.1(unstable)	1.000 / 1.000 / 1.000 / 1.000
1569	0.55	-0.279	0.018(stable); 0.027(unstable); 1.0(stable)	1.000 / 1.000 / 1.000 / 1.000
1569	1	-0.279	none (flows to boundary)	1.000 / 1.000 / 1.000 / 1.000
91569	0.3	2.319	0.999(stable)	0.178 / 0.656 / 1.000 / 1.000
91569	0.55	2.319	1.0(stable)	0.953 / 1.000 / 1.000 / 1.000
91569	1	2.319	1.0(stable)	1.000 / 1.000 / 1.000 / 1.000

qwen3:8b calibration of the 120 candidate claims found 4 SUPPORTS and 3 REFUTES claims with $|h_q| < 0.4$; a polarity-balanced set of 12 could not be formed, so genuine truth asymmetry remains unidentified in this model as well.

6.2 llama3.3:70b single-agent probe

claim	c_0	β_T	β_F	n	$P(\text{TRUE})$ at $k = 0$
6	1.987	0.636	-0.257	132	0.000
382	3.920	16.536	-0.500	132	0.000
1569	36.547	5.377	-4.482	132	0.000
2070	3.455	1.375	-0.554	132	0.000
90006	0.378	0.487	-0.494	132	0.417
90382	1.739	1.407	-0.748	132	1.000
91569	1.713	1.156	-0.572	132	1.000
92070	3.910	0.203	0.203	132	1.000

Polarity-coded regression across all eight claims (claim fixed effects): $b(\text{TRUE-side message}) = +0.583$, $b(\text{FALSE-side}) = -0.563$, ratio 1.03 (llama3.1:8b: 1.44). Per- k fits (truth-coded, pooled across claims): $\beta_T(4) = 2.41 \rightarrow \beta_T(8) = 1.35$. The positive wrong-side coefficient at $k = 4$ (+0.61) — wrong-side messages slightly *increasing* correct-side probability — is recorded as an observation only.

6.3 Recorded prediction misses

where	prediction	observation
llama3.1:8b, claim 1103, $\alpha=0.55$	wrong-side attractor ($x^*=0.305$)	correct-side convergence (0.67–0.79)
qwen3:8b, claim 1569 (all α)	wrong-side dominance	correct-side convergence ($\bar{x}_8 = 1.0$)
qwen3:8b, claim 91569, $\alpha=0.30$	correct-side monostable	threshold behavior at low x_0 ($\bar{x}_8 = 0.18$ at $x_0=4$)

7 Stage 2c: eight-claim calibration campaign and polarity-origin probe

7.1 Calibration campaign: all 16 cells

Predictions were frozen to `results/stage2c_blind_predictions.csv` (SHA-256 06d320c2...8572) about one minute before the collective driver launched. Observed crossings are 50% points of monotone logistic fits of episode outcomes against x_0 , with episode-bootstrap 95% intervals; $n_c/n_w/n_u$ are correct/wrong/undecided episode counts of 16 per cell.

7.2 Polarity-origin probe (D-3)

Judgment-side single-shot measurements on three llama3.1-8b variants with the fixed 8B message bank and a three-shot answer-format prefix applied identically to all variants. The base model does not follow the forced-choice format (parse rate 1.5% in the bare pilot, 4.9% under the prefix), so its row is reported for completeness and carries no statistical weight.

claim	α	predicted	pred. x_c	observed	obs. crossing [95% CI]	$n_c/n_w/n_u$	verdict
97	0.30	mono-correct	—	mono-correct	—	16/0/0	hit
97	0.55	mono-correct	—	mono-correct	—	16/0/0	hit
184	0.30	mono-correct	—	mono-correct	—	16/0/0	hit
184	0.55	mono-correct	—	mono-correct	—	16/0/0	hit
590	0.30	bistable(x-c=0.535)	0.535	bistable	0.563 [0.477, 0.657]	7/9/0	hit
590	0.55	bistable(x-c=0.629)	0.629	bistable	0.630 [0.527, 0.732]	6/10/0	hit
656	0.30	bistable(x-c=0.147)	0.147	bistable	0.246 [0.144, 0.362]	12/4/0	hit
656	0.55	bistable(x-c=0.125)	0.125	bistable	0.142 [0.097, 0.244]	14/2/0	hit
832	0.30	bistable(x-c=0.463)	0.463	bistable	0.507 [0.411, 0.608]	8/8/0	hit
832	0.55	bistable(x-c=0.325)	0.325	bistable	0.437 [0.352, 0.545]	9/7/0	hit
1013	0.30	bistable(x-c=0.491)	0.491	bistable	0.437 [0.352, 0.529]	9/7/0	hit
1013	0.55	bistable(x-c=0.415)	0.415	bistable	0.357 [0.328, 0.490]	10/6/0	hit
1146	0.30	bistable(x-c=0.129)	0.129	mono-correct	—	16/0/0	partial
1146	0.55	mono-wrong	—	bistable	0.063 [0.050, 0.164]	15/1/0	partial
1151	0.30	bistable(x-c=0.458)	0.458	bistable	0.188 [0.101, 0.333]	13/3/0	hit
1151	0.55	bistable(x-c=0.538)	0.538	bistable	0.313 [0.224, 0.416]	11/4/1	hit

Table 1: All 16 cells of the pre-registered calibration campaign (qwen3:8b, 256 episodes, zero parse failures). The two “partial” cells are claim 1146, whose predicted threshold (0.129 at $\alpha = 0.30$) lies at the edge of the seeding grid; the classification shifted by one category in each direction.

variant	tag (digest)	parsed	$ b_T / b_F $	95% CI
instruct (control)	llama3.1:8b (46e0c10c039e)	5,984/5,984	1.437	[1.250, 1.651]
abliterated	mannix/llama3.1-8b-abliterated (f7d729db9832, Q4_0)	5,984/5,984	1.243	[1.051, 1.471]
base	llama3.1:8b-text-q4_K_M (6f98b5a6e4b7)	294/5,984	(2.10)	not evaluable

Table 2: Per-message assertion asymmetry (claim-fixed-effect logistic regression; claim-clustered bootstrap intervals). The instruct control reproduces the Stage-1 value 1.44 under the added format prefix. Bare affirmation indices S for the four original/negated pairs: instruct 0.92–1.52 (mean 1.22), abliterated 1.13–1.42 (mean 1.24); base cells parsed 2–6 of 64 and are not evaluable.

8 Residual and consistency analyses

8.1 Message statistics by side

claim	side	messages	words/msg	hedges/msg
6	correct	28258	60.7	0.058
6	wrong	35998	59.4	0.419
382	correct	16713	63.5	0.035
382	wrong	47591	62.9	0.116
1569	correct	28115	58.8	0.084
1569	wrong	36093	59.6	0.314
2070	correct	14105	64.5	0.081
2070	wrong	50119	59.8	1.494

8.2 Recorded-message persuasiveness probes

claim	side	message source	$P(\text{correct})$	n
6	correct	bank	1.000	60
6	correct	episode	1.000	60
6	wrong	bank	0.100	60
6	wrong	episode	0.167	60
382	correct	bank	1.000	60
382	correct	episode	1.000	60
382	wrong	bank	0.017	60
382	wrong	episode	0.000	60
1569	correct	bank	1.000	60
1569	correct	episode	1.000	60
1569	wrong	bank	0.100	60
1569	wrong	episode	0.050	60
2070	correct	bank	1.000	60
2070	correct	episode	1.000	60
2070	wrong	bank	0.050	60
2070	wrong	episode	0.167	60

Episode-generated and bank messages are indistinguishable in persuasive force; the high- x_0 residual is not a message-quality effect.

8.3 Repeated exposure and finite time

One-step regression with round interactions gives $\beta_T \times (t-1) = +0.026$ (SE 0.006) and $\beta_F \times (t-1) = -0.004$ (SE 0.003): repeated exposure does not attenuate message influence. Finite-time check on high- x_0 episodes:

α	$32x_0$	n	mean $x_8 - x_7$	frac. interior	frac. still moving
0.2	20	40	-0.008	0.300	0.275
0.2	24	40	-0.009	0.225	0.150
0.3	20	40	-0.012	0.450	0.250
0.3	24	40	+0.001	0.275	0.325
0.38	20	40	+0.001	0.450	0.375
0.38	24	40	-0.028	0.375	0.350
0.45	20	16	-0.020	0.438	0.375
0.45	24	40	-0.008	0.275	0.375
0.55	20	40	-0.009	0.525	0.375
0.55	24	40	-0.002	0.450	0.225
1	20	40	-0.002	0.800	0.575
1	24	40	-0.027	0.725	0.550

8.4 Label residual

Correlation between an episode’s share of TF label assignments and its final state: $r = 0.194$ ($n = 1,008$ episodes) — an order of magnitude below the experimental effects ($\Delta \bar{x}_8 \approx 1$), but nonzero; per-call randomization protects pooled estimates.

8.5 Flip statistics and cascades

Refractory regression: the coefficient of “flipped at $t-1$ ” on flipping at t is $+0.98$ (claim-cluster SE 0.13) — flips concentrate in particular agents (anti-refractory). Branching ratios:

α	flip events	σ (mean downstream flips)
0.2	9048	3.546
0.3	10129	3.102
0.38	11452	2.771
0.45	10473	2.476
0.55	12301	2.312
1	14625	1.776

Cascade-size distribution in the asynchronous public dataset (GPT-4o-mini, 320 runs; cascades linked by graph adjacency within three events):

cascade size	count
1	9341
2	668
3	130
4	28
5	9
6	1
7	1

91.8% of cascades have size 1; the tail decays geometrically (deeply subcritical).

9 Fidelity check

Ten episodes at $\alpha = 0.30$, $x_0 = 16/32$ re-run with five-sample majority voting: wrong-consensus rate 1.00, undecided 0.00, mean $|x_8 - 0.5| = 0.44$, versus 0.875, 0.125, 0.38 under single-sample judgments — absorption is sharper, not weaker, under the original five-sample protocol.